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ON THE GLOBAL INSTABILITY OF SQUARE AND CROSS LID-DRIVEN CAVITY FLOWS WITH A PURE STREAMFUNCTION-VELOCITY (Ψ-V) APPROACH

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ABSTRACT
This manuscript embodies an investigation of flow stabilities in lid-driven cavities of square (one-sided) and cross (two-sided) geometry with the scheme developed by Kalita and Gupta, which was derived by using the pure streamfunction-velocity (ψ-v) formulation of the 2-D incompressible Navier-Stokes' equations. In the current work, for both the geometries, the authors, for the first time globally analyse the flow stability utilizing the cited scheme which was based on pure streamfunction-velocity formulation. They also perform a validation study of the applied scheme on the two-sided cross lid-driven cavity flow through a comparison exercise for a a certain range of Reynolds numbers and excellent comparison with established results have been obtained. Critical parameters have been studied with the help of eigenvalue analysis and results have been reported. It has been observed that there is a reasonable agreement with established results for the square cavity, while for the two-sided cross cavity, it's global stability analysis has been carried out for the first time and new results are obtained.

Keywords: lid-driven, geometry, global stability, navier-stokes, higher order compact, accuracy.

1. INTRODUCTION

Dynamics of fluid flows in lid-driven cavities (LDC) over the last few decades has been most extensively studied. It has been the most widely used benchmark problem for the assessment of numerical algorithms developed for fluid motion governed by the Navier-Stokes equations. Literatures suggest that the global stability of the one-sided square lid-driven cavity has been analysed by numerous researchers and various techniques have been employed with several established numerical methods [2-8].

The present work intends to obtain accurate values of the critical Reynolds number (Re_c). Existing literatures reveal that the Re_c obtained with different techniques could be quite different from each other. A comprehensive study reveals that Re_c value lie in the range 7704-8031.93, for the square cavity [2,7]. In contrast to this, the global stability analysis of the two-sided cross LDC has been carried out for the first time and new results are observed.

In the current study, the authors utilized the scheme [1] proposed by Kalita and Gupta. This scheme was developed by using the pure streamfunction-velocity (ψ-v) formulation of the 2-D incompressible Navier Stokes equations. The scheme has been used to simulate the steady state flow and then the same scheme has been utilized to analyse the global stability of the fluid flow by solving the two-dimensional generalized eigenvalue problem (GEP) obtained from the linearized unsteady equations.

Prior to carrying out the global stability analysis, the authors examine the validity of the mentioned scheme by performing a validation study of the scheme on the two-sided cross LDC flow through a comparison exercise for a certain range of Reynolds numbers and compare their results with those of Vicente et al. [8]. Excellent comparison has been observed.

The paper has been organized in six sections. In section 2, the authors describe the problem and governing equations, section 3 deals with the normal mode analysis, section 4 the numerical scheme and other related issues, section 5 the global stability analysis and section 6 summarizes the whole work.

2. PROBLEM DESCRIPTION

Two problems of different geometries have been considered here. The configurations have been shown in Fig.1. For the square LDC, the top lid is moving

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